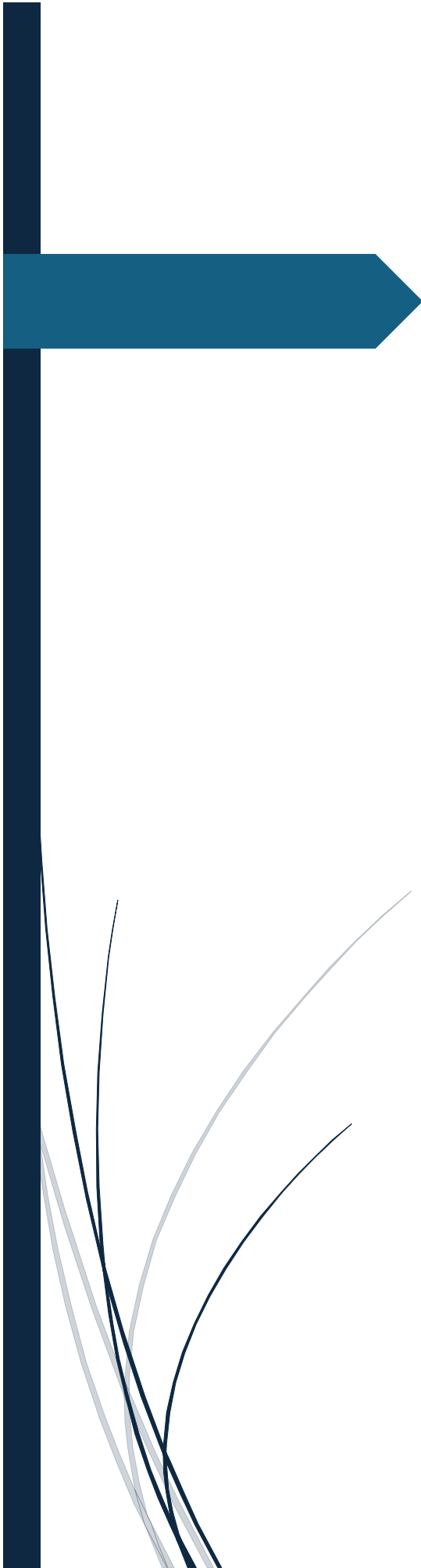




Assignment 5

Case Study Practical

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

❖ Laptop Won't Power On: Troubleshooting Steps

- **First, check the power outlet** → Make sure the wall socket is working properly.
- **Then, check the charger** → Check the charger, cable, and charging indicator for any damage or problem.
- **Then, check the charging port** → Make sure the charger is connected firmly and the port is not loose or damaged.
- **Next, perform a power reset** → Disconnect the charger, hold the power button for 15–30 seconds, then reconnect the charger and try again.
- **Finally, check the battery** → If possible, test with another compatible charger or battery to determine whether the battery is faulty.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue. Hint: Consider checking both hardware connections and basic display settings.

❖ Blank Monitor Troubleshooting:

- 1. Check the monitor power** → Make sure the monitor is switched on and the power cable is connected.
- 2. Check the display cable** → Check the HDMI/DisplayPort/VGA cable at both ends.
- 3. Check the input source** → Make sure the monitor is set to the correct HDMI/DisplayPort input.
- 4. Check the GPU connection** → Make sure the graphics card is properly connected and powered.
- 5. Restart the PC** → Check if the display appears after restarting.
- 6. Finally, check display settings** → Press **Windows + P** and select **PC screen only** or **Duplicate**.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings

❖ Wireless Mouse Troubleshooting:

- 1. First, check the mouse** → Replace the battery and make sure the mouse is switched on.
- 2. Then, check the USB receiver** → Remove and reconnect the receiver to the USB port.
- 3. Next, try another USB port** → If it works, the original USB port may be faulty.
- 4. Then, test another mouse** → If another mouse works, the original mouse is the problem.
- 5. Finally, check system settings** → Check Device Manager and reinstall/update the mouse driver.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one. Hint: Think about fans, dust, and thermal paste.

❖ Overheating and Shutdown: Possible Causes:

- 1. Dust buildup** → Check the fans, vents, and heatsink for dust. Clean them if necessary.
- 2. Faulty/slow fan** → Check whether the CPU/GPU fans are spinning properly during use.
- 3. Old thermal paste** → Check the thermal paste between the CPU/GPU and heatsink. Replace it if it is dry or damaged.